MATH 345 Skeleton Notes

Unit 5: Eigenvalues and second-order optimization

Section 5.2: Symmetric matrices and orthogonal diagonalization

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A symmetric matrix, as defined in Symmetric matrices, is especially well behaved. Its eigenvectors can be chosen to form an orthonormal basis; recall Orthogonal and orthonormal sets. Thus diagonalization uses a rigid change of coordinates that preserves dot products and lengths. This section develops that structure before applying it to quadratic forms.

Orthogonality of eigenvectors, orthogonal matrices, and orthogonal diagonalization

A key property of symmetric matrices is that for any two vectors $\mathbf{x}$ and $\mathbf{y}$, $(A\mathbf{x}) \cdot \mathbf{y} = \mathbf{x} \cdot (A\mathbf{y})$. Indeed, we calculate that

$$\begin{aligned}(A\mathbf{x}) \cdot \mathbf{y} &= (A\mathbf{x})^T \mathbf{y} \\ &= \mathbf{x}^T A^T \mathbf{y} \\ &= \mathbf{x}^T A\mathbf{y} \quad (\text{since } A^T = A) \\ &= \mathbf{x} \cdot (A\mathbf{y})\end{aligned}$$

This property leads to a fundamental result about the eigenvectors of symmetric matrices.

Theorem: Orthogonality of Eigenvectors

If A is a *symmetric matrix*, then the *eigenvectors* of A corresponding to *distinct* eigenvalues are *orthogonal*.

Why is this true?

Let $A\mathbf{x} = \lambda\mathbf{x}$ and $A\mathbf{y} = \mu\mathbf{y}$, where $\lambda \neq \mu$. Using the property above:

$$\begin{aligned}\lambda(\mathbf{x} \cdot \mathbf{y}) &= (\lambda\mathbf{x}) \cdot \mathbf{y} \\ &= (A\mathbf{x}) \cdot \mathbf{y} \\ &= \mathbf{x} \cdot (A\mathbf{y}) \\ &= \mathbf{x} \cdot (\mu\mathbf{y}) \\ &= \mu(\mathbf{x} \cdot \mathbf{y})\end{aligned}$$

This gives us $(\lambda - \mu)(\mathbf{x} \cdot \mathbf{y}) = 0$. Since $\lambda \neq \mu$, we must have $\mathbf{x} \cdot \mathbf{y} = 0$, which means the eigenvectors are orthogonal.

As we will see, it turns out that for any symmetric matrix, there exists a pairwise orthogonal set of eigenvectors which form a basis of $\mathbb{R}^n$. In particular, every symmetric matrix is diagonalizable.
Recall from Orthogonal Matrices that a square matrix Q is *orthogonal* if $Q^T = Q^{-1}$.

Theorem: Equivalent Conditions for Orthogonal Matrices

Let Q be an $n \times n$ matrix. Then the following conditions are equivalent to one another:

1. Q is invertible and $Q^{-1} = Q^T$.
2. The rows of Q are orthonormal.
3. The columns of Q are orthonormal.

Why is this true?

Let us start by showing the first property is equivalent to the third. Suppose Q is invertible, and $Q^{-1} = Q^T$. Let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be the columns of Q. Recall from Matrix multiplication by row-column dot products that the (i, j)-entry of Q^TQ is equal to the dot product $\mathbf{v}_i \cdot \mathbf{v}_j$. Since $Q^{-1} = Q^T$, $Q^TQ = I$, and so for $i \neq j$, $\mathbf{v}_i \cdot \mathbf{v}_j = 0$, and for each i, $\mathbf{v}_i \cdot \mathbf{v}_i = 1$. Thus the

column vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are orthonormal. Conversely, if the column vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are orthonormal, then we immediately see that $Q^T Q = I$. The proof that the first property is equivalent to the second is very similar, using the rows of Q rather than the columns and that if $Q^T Q = I$, then $QQ^T = I$, and vice versa.

Lemma

Let Q be an $n \times n$ orthogonal matrix. Then:

1. Q preserves dot products, i.e., for any vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, $(Q\mathbf{v}) \cdot (Q\mathbf{w}) = \mathbf{v} \cdot \mathbf{w}$.
2. Q preserves lengths, i.e., for any vector $\mathbf{v} \in \mathbb{R}^n$, $\|Q\mathbf{v}\| = \|\mathbf{v}\|$.

Why is this true?

Let $\mathbf{v}$ and $\mathbf{w}$ be column vectors. Recalling the fifth property of Properties of matrix multiplication, we calculate that

$$\begin{aligned}(Q\mathbf{v}) \cdot (Q\mathbf{w}) &= (Q\mathbf{v})^T (Q\mathbf{w}) \\ &= \mathbf{v}^T Q^T Q\mathbf{w} \\ &= \mathbf{v}^T I_n \mathbf{w} \\ &= \mathbf{v}^T \mathbf{w} \\ &= \mathbf{v} \cdot \mathbf{w}.\end{aligned}$$

Thus Q preserves dot products.
Next, using that Q preserves dot products, and also the referenced item, we find

$$\|Q\mathbf{v}\|^2 = (Q\mathbf{v}) \cdot (Q\mathbf{v}) = \mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2.$$

Taking square roots on both sides of this equation gives $\|Q\mathbf{v}\| = \|\mathbf{v}\|$.

The figure provides a visualization of an orthogonal change of coordinates.

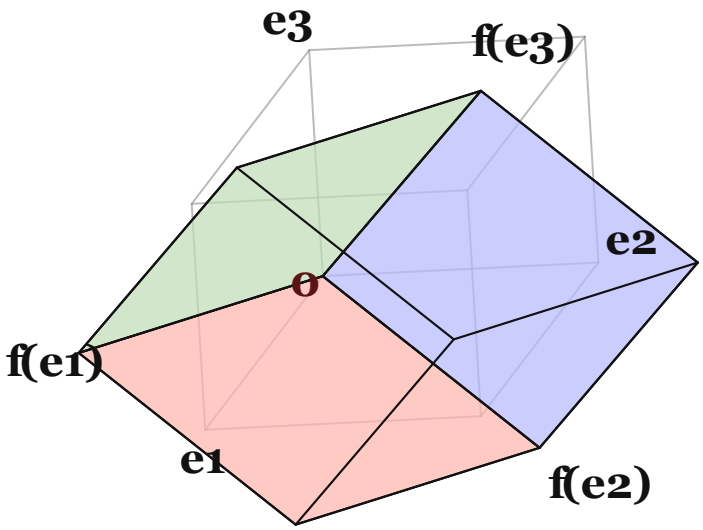


Figure. A cube and its rotated image under an orthogonal matrix.

Definition: Orthogonal diagonalizability

A matrix A is orthogonally diagonalizable when an *orthogonal matrix* Q can be found such that $Q^{-1}AQ$ is a diagonal matrix. Since $Q^{-1} = Q^T$, this is equivalent to $Q^T AQ$ being a diagonal matrix.

Activity: Checking eigenvectors and orthogonality (U5-LO1, U4-LO1)

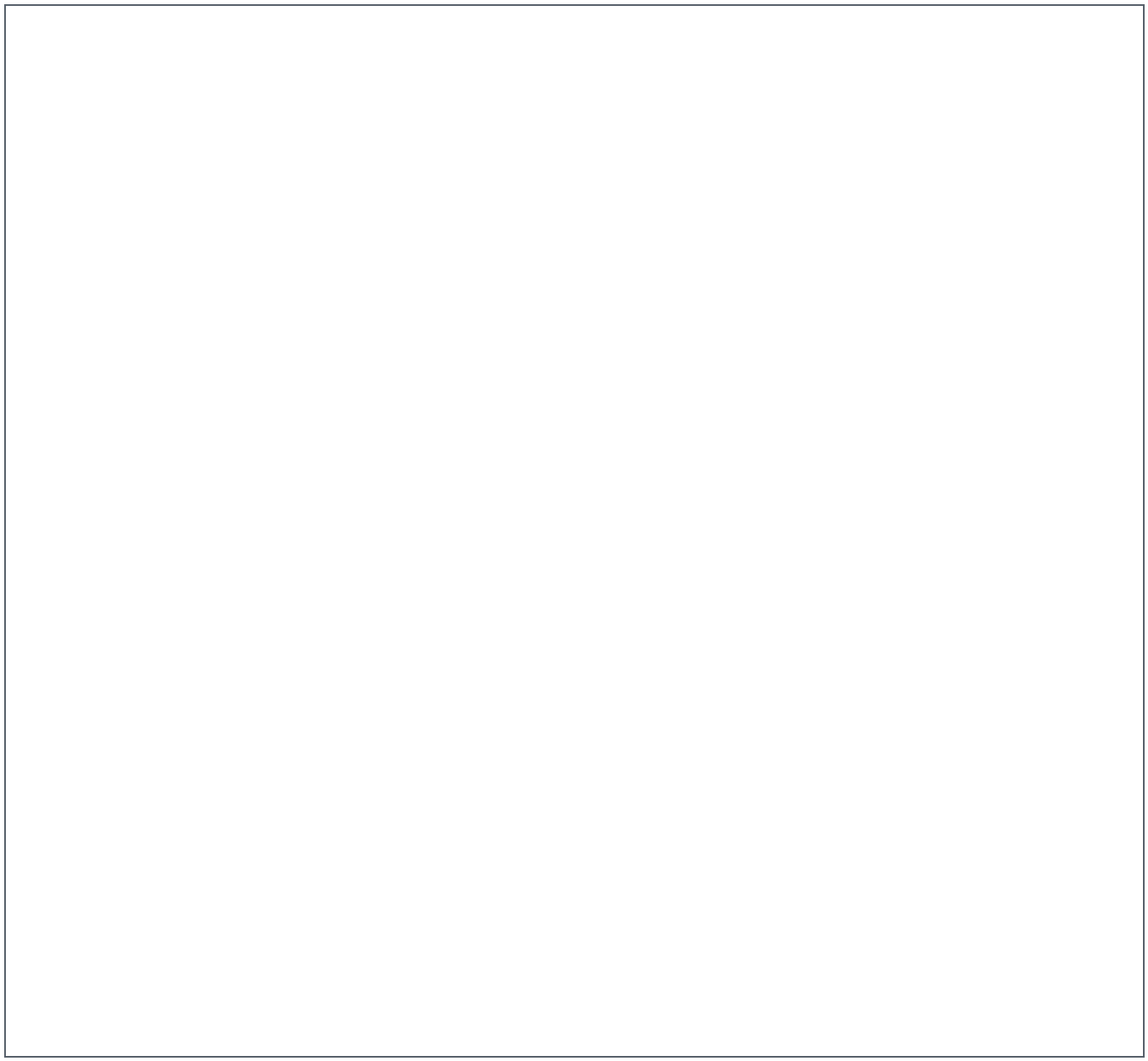
Consider the symmetric matrix

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Check that the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

are eigenvectors of A which are orthogonal to each other. What are the corresponding eigenvalues?



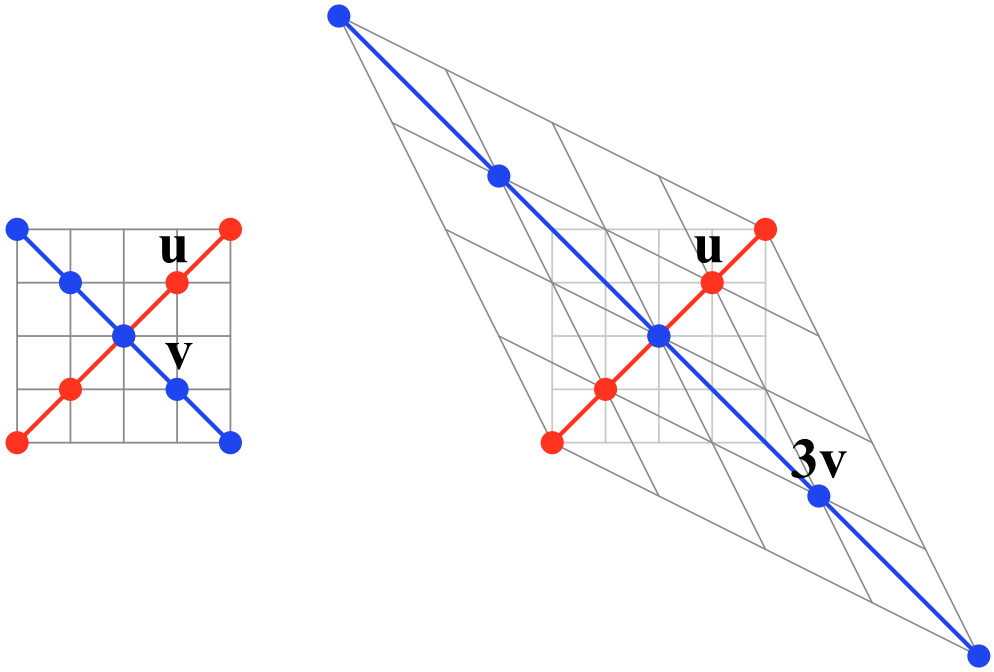


Figure. Before-and-after grid diagram for a symmetric matrix with orthogonal eigenvector directions.

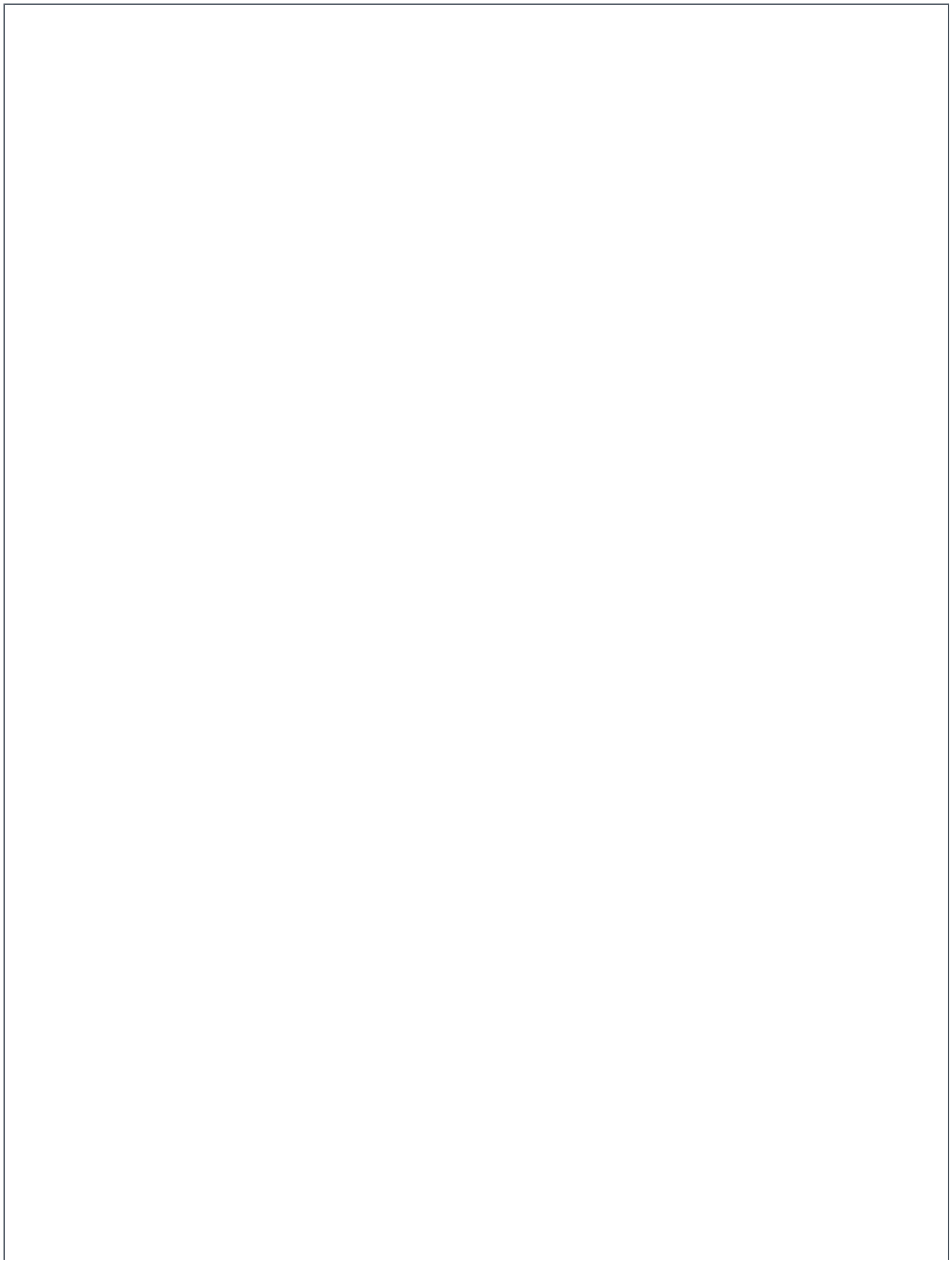
The figure should be compared to Before-and-after grid diagram for a matrix with two highlighted eigenvector directions. Notice that since the matrix A in Checking eigenvectors and orthogonality (U5-LO1, U4-LO1) is symmetric, the red and blue line in the figure meet at right angles, since the vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, whereas the red and blue lines in Before-and-after grid diagram for a matrix with two highlighted eigenvector directions do not meet at right angles, since the eigenvectors for that example are not orthogonal to one another.

Activity: Orthogonal diagonalization in two dimensions (U5-LO3)

Consider the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Find an orthogonal matrix Q such that $Q^{-1}AQ$ is diagonal.



Activity: Rewriting an orthogonal diagonalization (U5-LO3)

Suppose that A is orthogonally diagonalizable, i.e., we can find an orthogonal matrix Q so that $Q^T AQ = D$, where D is diagonal. Explain why $A = QDQ^T$.

Remark

The factorization $A = QDQ^T$ writes A as a product of three matrices: an orthogonal matrix, a diagonal matrix, and the transpose of the orthogonal matrix. This product is sometimes called

the spectral decomposition of a symmetric matrix A.

Principal axes theorem

Recall Diagonalizability criterion. That theorem tells us that if A is orthogonally diagonalizable, where $Q^{-1}AQ$ is diagonal, then the columns of Q must be linearly independent eigenvectors of A. In addition, in order for Q to be orthogonal, Equivalent Conditions for Orthogonal Matrices tells us that the eigenvectors which form the columns of Q must be orthonormal.

Remarkably, *all* symmetric matrices are orthogonally diagonalizable.

Theorem: Principal Axes Theorem

Let A be any $n \times n$ matrix. Then the following conditions are equivalent:

1. A has an orthonormal set of n eigenvectors
2. A is orthogonally diagonalizable
3. A is symmetric

A set of orthonormal eigenvectors for a symmetric matrix A is called a set of principal axes for A.

Why is this true?

We do not prove the "hard" part of the theorem (that the third property implies the second). But the proof that the second property implies the third is simple (see Orthogonal diagonalization and symmetry (U5-LO3)).

Remark

Principal Axes Theorem is also called The Spectral Theorem for symmetric matrices. The set of eigenvalues of a matrix is often called the spectrum of the matrix.

We do not prove the "hard" part of this theorem, that is, that (3) implies (2). Let's however check that (2) implies (3).

Activity: Orthogonal diagonalization and symmetry (U5-LO3)

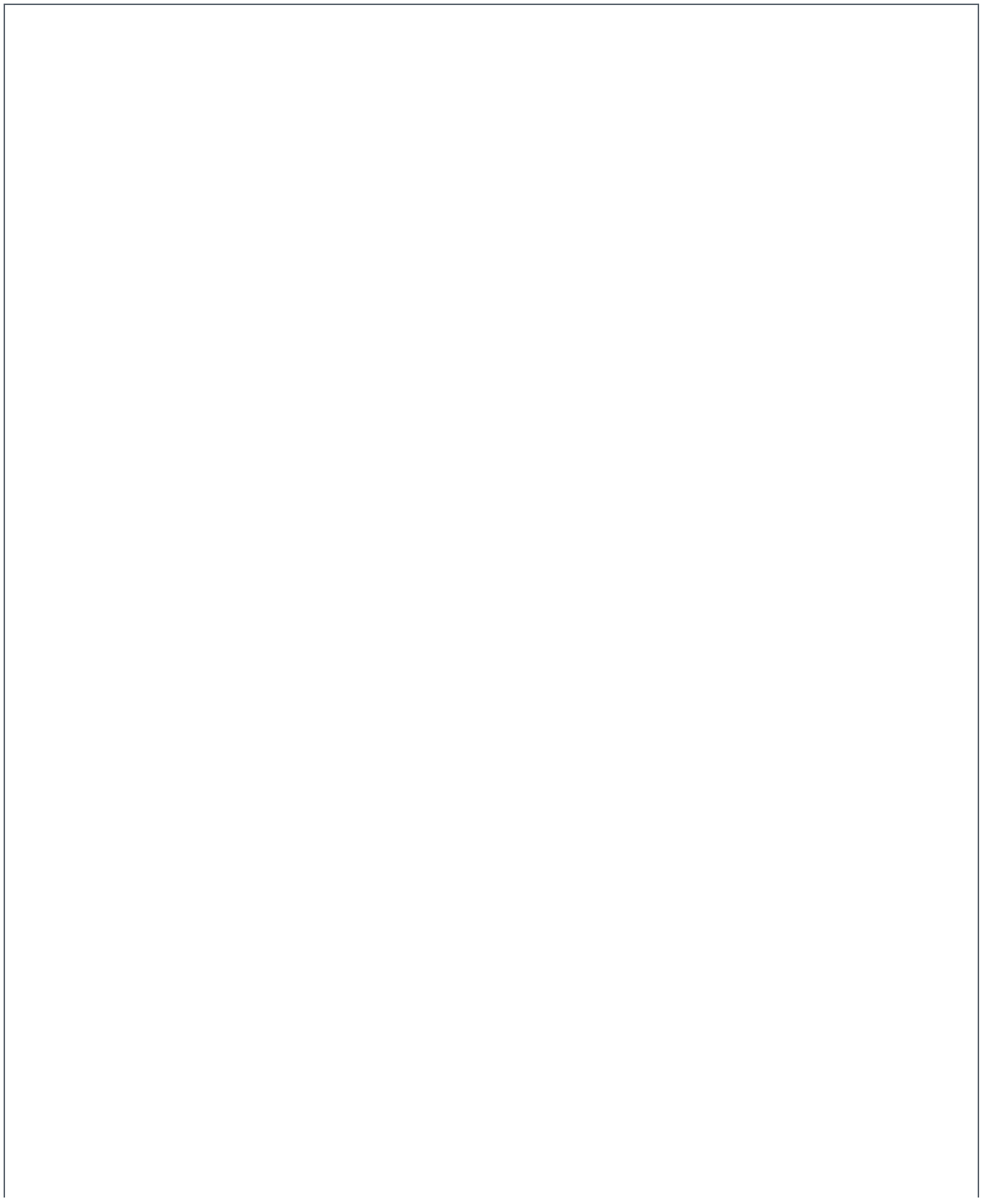
Show that if A is orthogonally diagonalizable, then A is symmetric.

Activity: Orthogonal diagonalization in three dimensions (U5-LO3)

Let

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ -1 & 2 & 5 \end{bmatrix}$$

Find an orthogonal matrix Q such that $Q^{-1}AQ$ is diagonal.



In Orthogonal diagonalization in three dimensions (U5-LO3), the eigenvalues of A were distinct. For a general strategy, recall that the eigenspace $E_\lambda(A)$ is the set of vectors $\mathbf{x}$ satisfying $A\mathbf{x} = \lambda\mathbf{x}$. It is a subspace of $\mathbb{R}^n$ by Eigenspaces are subspaces, so we can find an orthonormal basis for it.

Note: Procedure for orthogonally diagonalizing a symmetric matrix A

1. Find the eigenvalues of A.
2. For each eigenvalue λ of A, find a basis for $E_\lambda(A)$, and convert it to an orthonormal basis using the Gram-Schmidt orthogonalization process (see Properties of the Gram-Schmidt output) if necessary.
3. Collecting the orthonormal bases for each of the eigenspaces gives an orthonormal basis of eigenvectors for A. Taking these eigenvectors as the columns of a matrix Q gives an orthogonal matrix such that $Q^{-1}AQ$ is diagonal.

Activity: Orthogonal diagonalization from eigenspace data (U5-LO3)

If

$$A = \begin{bmatrix} 8 & -2 & 2 \\ -2 & 5 & 4 \\ 2 & 4 & 5 \end{bmatrix},$$

compute an orthonormal diagonalization of A. You may use the fact that the characteristic polynomial of A is $\lambda(\lambda - 9)^2$, and the null space of A is spanned by the vector

$$\begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}.$$

